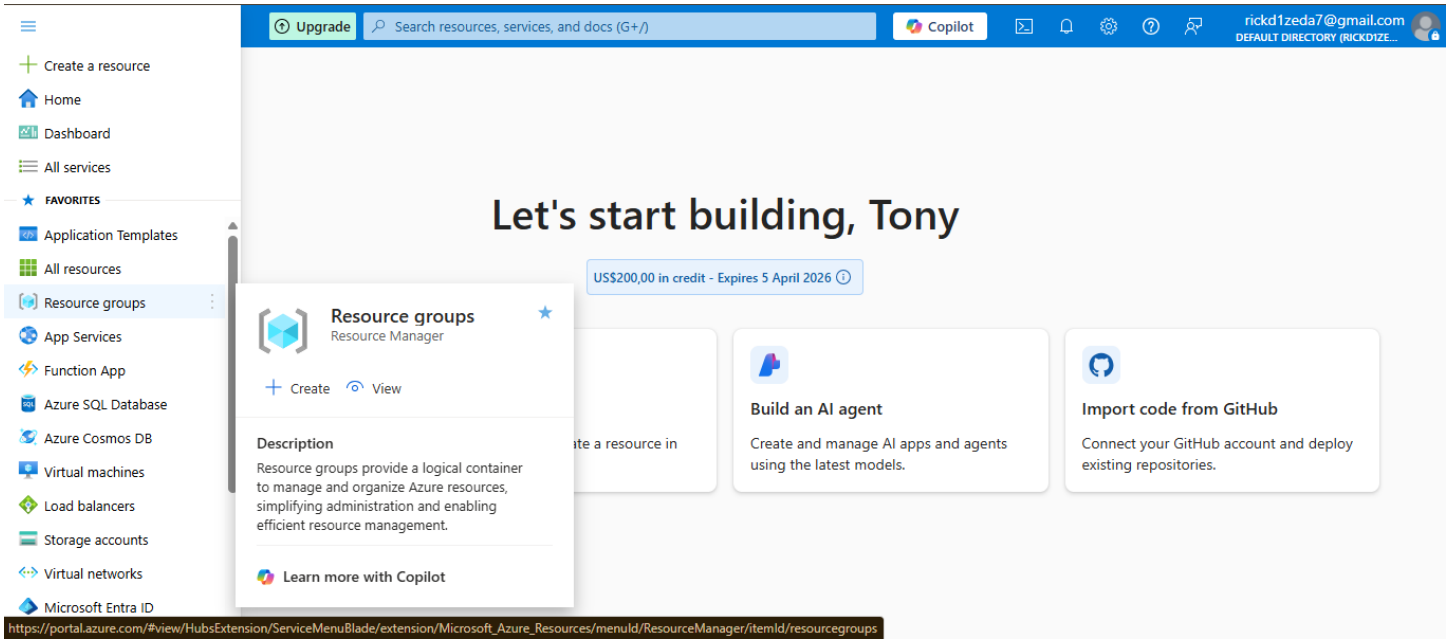
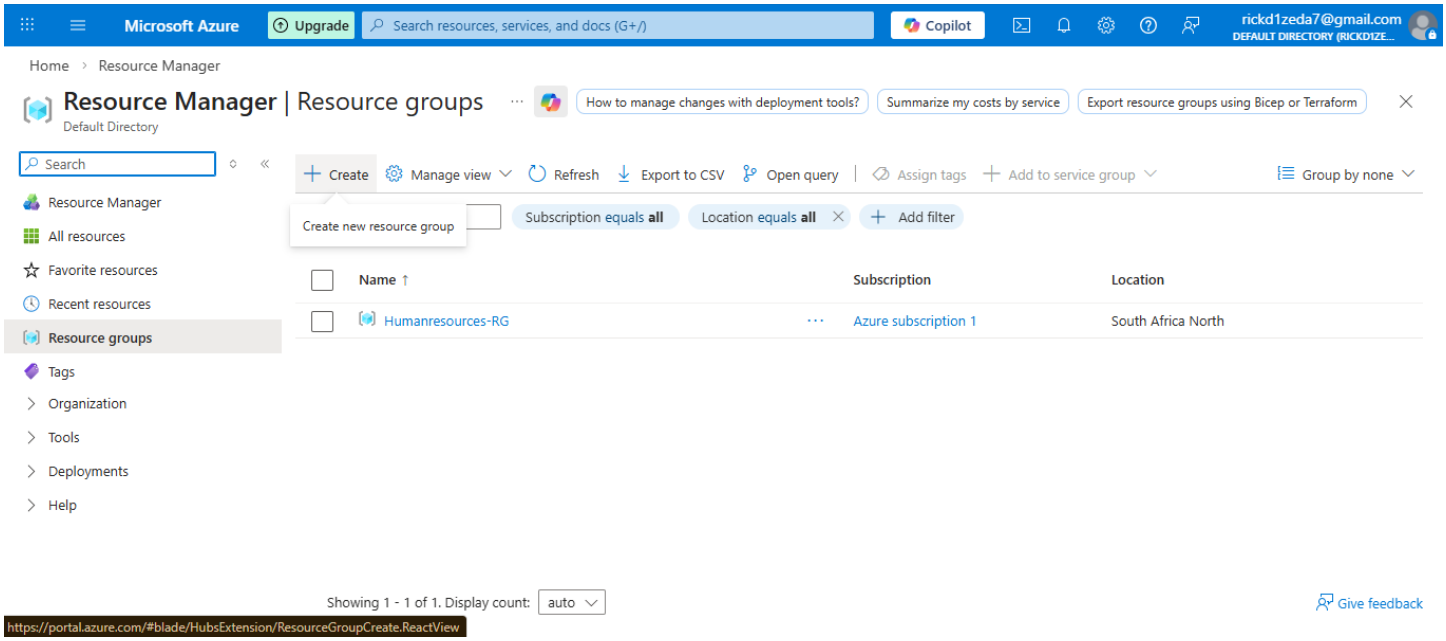
Mthunzelwa Makwanyane
20241063

# Practical labs (Week 2)

Lab activity 1: Creating a resource group



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Home > Resource Manager | Resource groups

## Create a resource group

Basics | Tags | Review + create

**Resource group** - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. Learn more

Subscription * : Azure subscription 1

Resource group name * : Developers-RG

Region * : (Africa) South Africa North

Previous | Next | Review + create

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Home > Resource Manager | Resource groups

## Create a resource group

Basics | Tags | Review + create

Automation Link

**Basics**

| | |
|---|---|
| Subscription | Azure subscription 1 |
| Resource group name | Developers-RG |
| Region | South Africa North |

**Tags**

None

Previous | Next | Create

Home > Resource Manager

## Resource Manager | Resource groups

Default Directory

Search

- Resource Manager
- All resources
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- Resource groups
- Tags
- Organization
- Tools
- Deployments
- Help

Create | Manage view | Refresh | Group by none

Filter for any field...

Subscription equals all | Location equals all | Add filter

| Name ↑ | Subscripti |
|---|---|
| Humanresources-RG | Azure sub: |

Showing 1 - 1 of 1. Display count: auto

Give feedback

Add or remove favorites by pressing Ctrl+Shift+F

## Notifications

More events in the activity log → | Dismiss all

**Resource group created**

Creating resource group 'Developers-RG' in subscription 'Azure subscription 1' succeeded.

Go to resource group | Pin to dashboard

a few seconds ago

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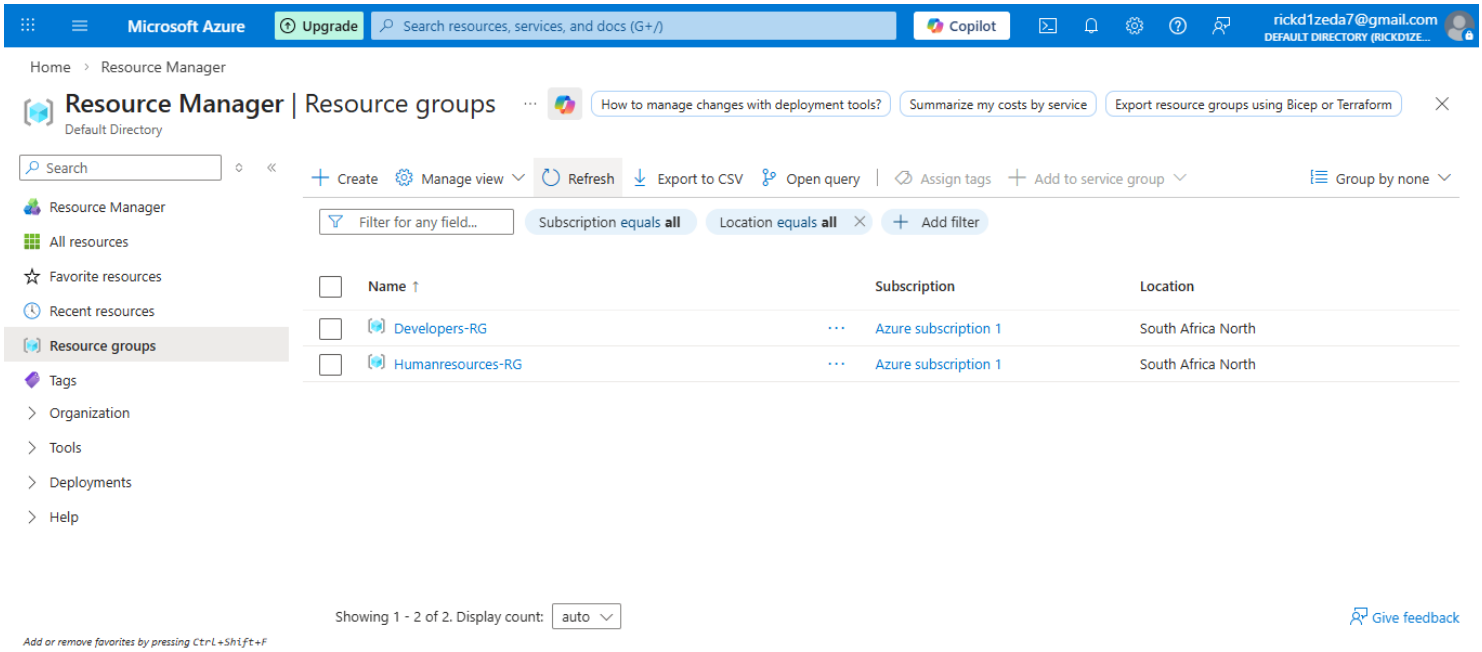
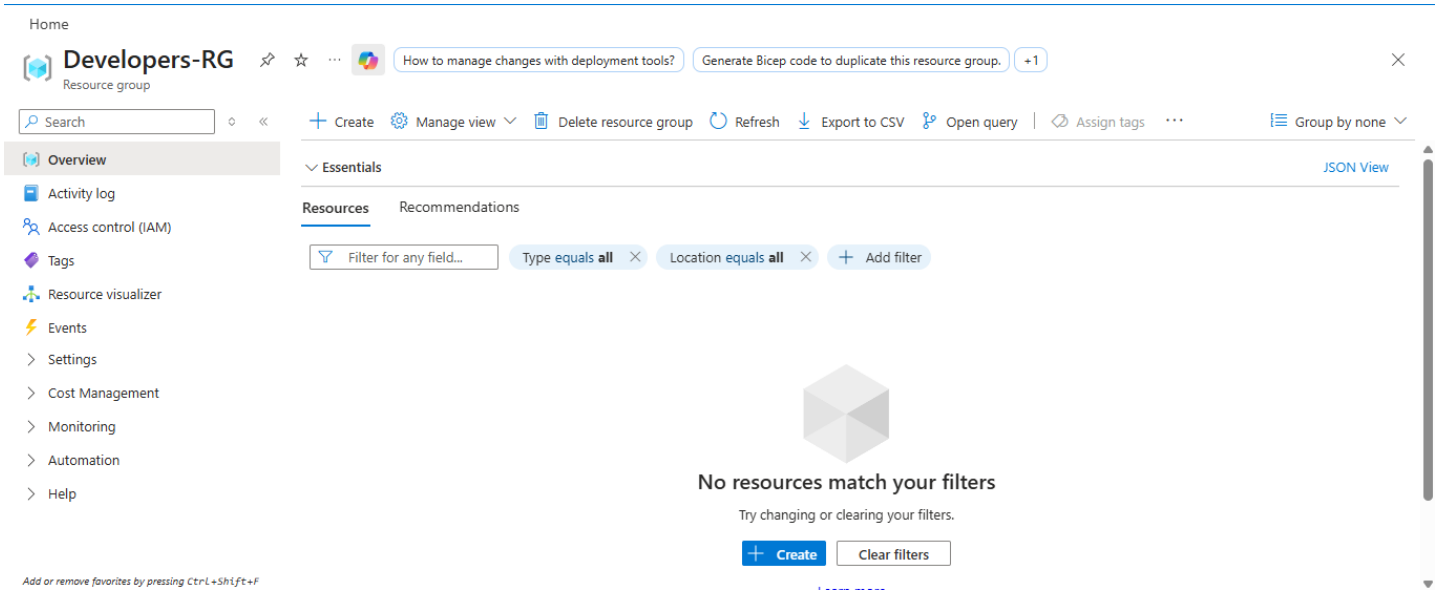
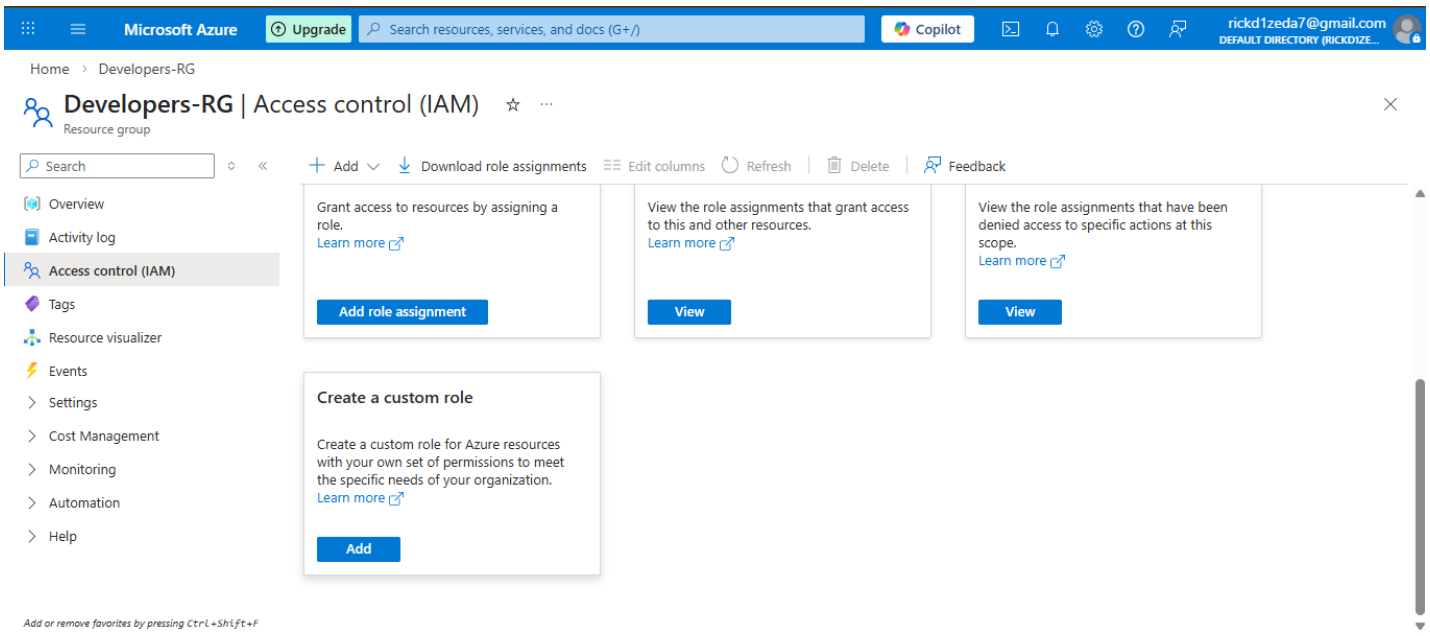



## Lab activity 2: Creating a custom role-based access control



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Home > Developers-RG | Access control (IAM)

## Create a custom role

Basics | Permissions | Assignable scopes | JSON | Review + create

To create a custom role for Azure resources, fill out some basic information. Learn more

Custom role name * cannot remove content

Description Users in this resource group can add data but not not remove it

Baseline permissions ○ Clone a role ◉ Start from scratch ○ Start from JSON

Review + create | Previous | Next | Feedback

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Home > Developers-RG | Access control (IAM)

## Create a custom role

Basics | Permissions | Assignable scopes | JSON | Review + create

+ Add permissions | + Exclude permissions

Click Add permissions to select the permissions you want to add to this custom role.
To add a wildcard (*) permission, you must manually add the permission on the JSON tab. Learn more
To exclude specific permissions from a wildcard permission, click Exclude permissions. Learn more

| Permission | Description | Permission type |
|---|---|---|
| No permissions to display. | | |

Add permissions

### Definitions

**Control plane**
Actions specify the operations that a role is allowed to perform. NotActions specify the operations that are excluded from the allowed Actions (this is useful if a role has wildcards).

**Data plane**
DataActions specify the operations that a role is allowed to perform to the data within an object. NotDataActions specify the operations that are excluded from the allowed DataActions (this is useful if a role has wildcards).

Review + create | Previous | Next | Feedback

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Home > Developers-RG | Access

## Create a custom role

Basics | Permissions | Assign

+ Add permissions | + Exclude

Click Add permissions to select th
To add a wildcard (*) permission, y
To exclude specific permissions fr

| Permission |
|---|
| No permissions to display. |

Review + create | Previou

## Add permissions

**Microsoft Data Factory**
Hybrid data integration at enterprise scale, made easy.

**Microsoft Data Replication**
Microsoft.DataReplication

**Microsoft Data Share**
A simple and safe service for sharing big data with external organizations.

**Microsoft Databricks**
Fast, easy, and collaborative Apache Spark-based analytics platform.

**Microsoft DataLakeAnalytics**
Distributed analytics service that makes big data easy.

**Microsoft DataLakeStore**
Highly scalable and cost-effective data lake solution for big data analytics.

**Microsoft DB for MariaDB**
Managed MariaDB database service for app developers.

**Microsoft DB for MySQL**
Managed MySQL database service for app developers.

**Microsoft DB for PostgreSQL**
Managed PostgreSQL database service for app developers.

**Microsoft Developer Hub**
Microsoft.DevHub

**Microsoft Devices**
Ensure that your users are accessing your resources from devices that meet your standards for security and compliance.

**Microsoft DevTest Labs**
Quickly create environments using reusable templates and artifacts.

**Microsoft DocumentDB**

**Microsoft Domains**

**Microsoft EnterpriseSupport**

**Microsoft Event Grid**

Add | Cancel | Download all permissions

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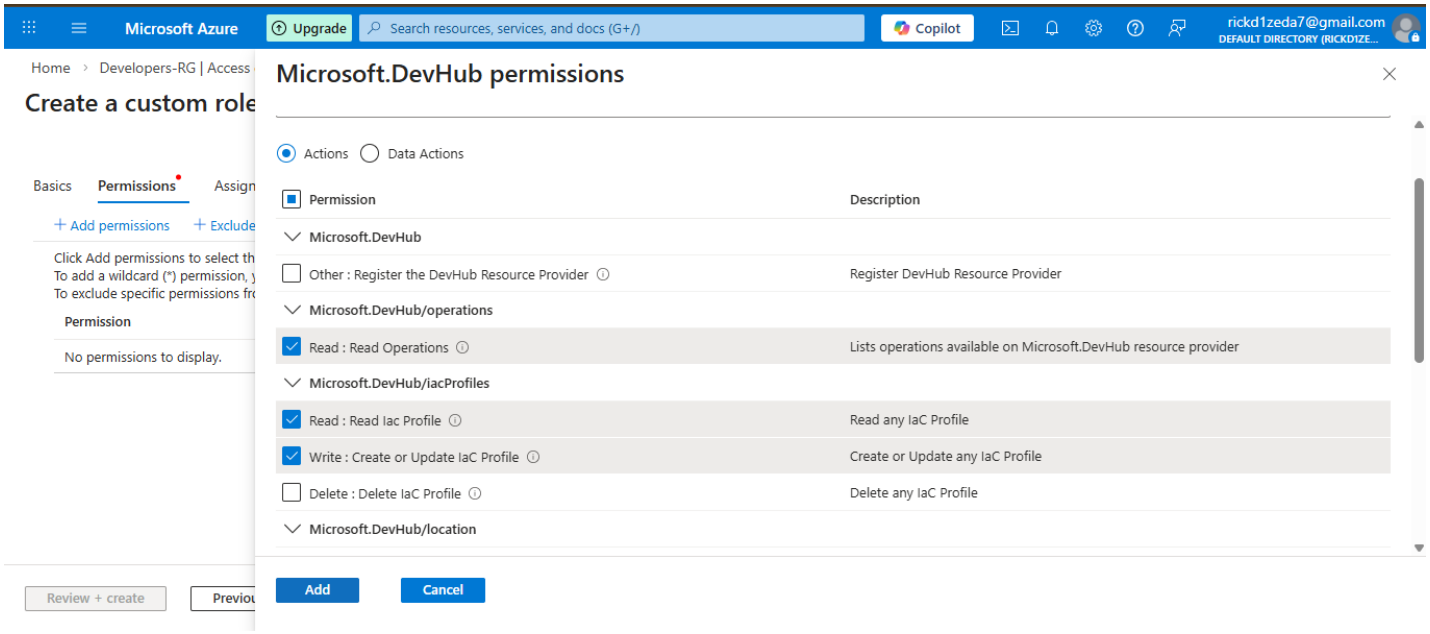


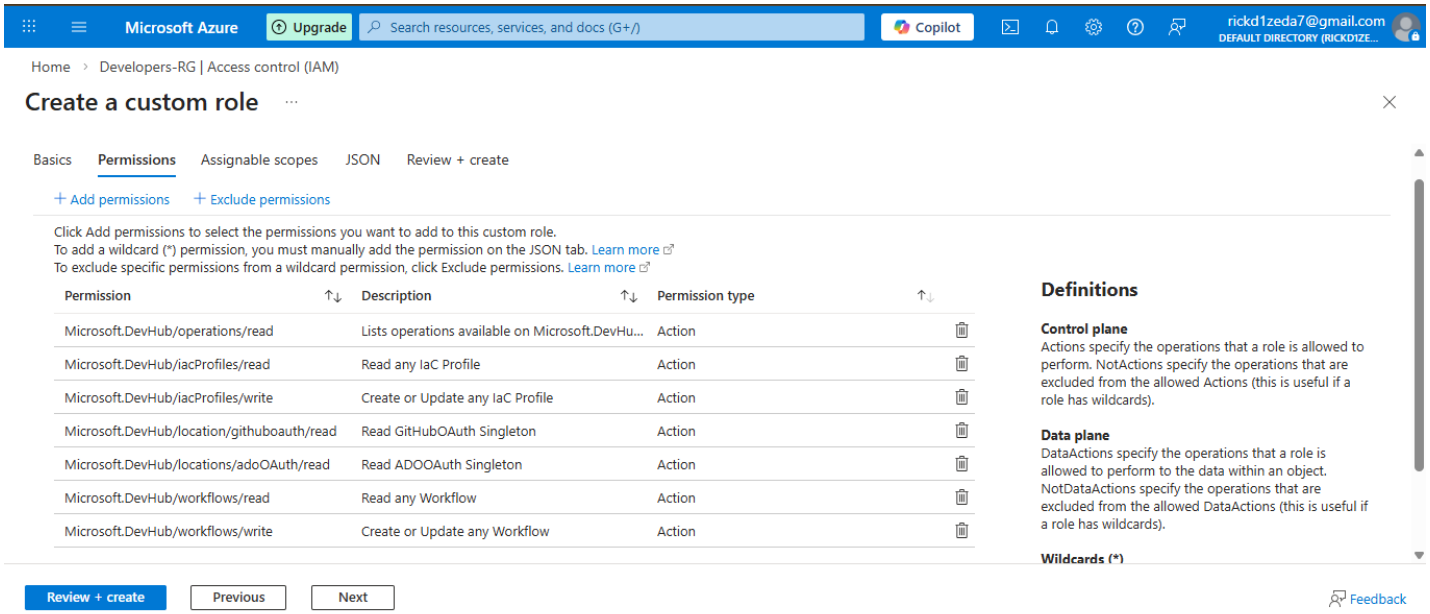


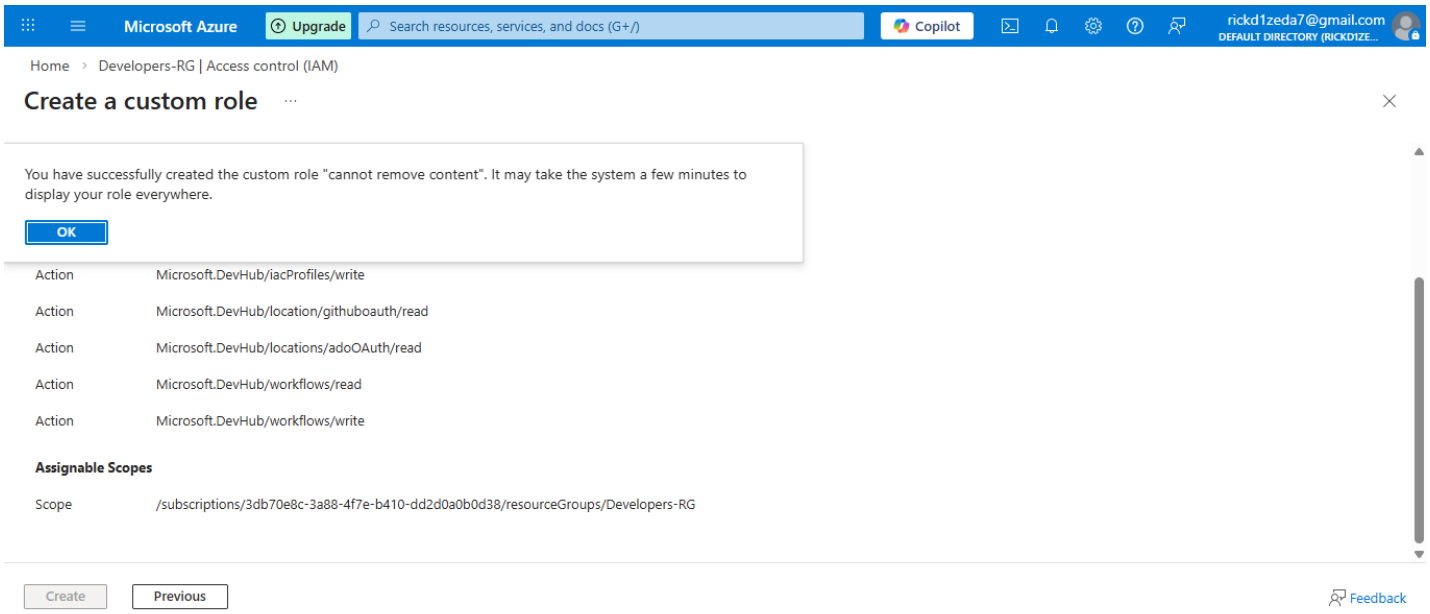

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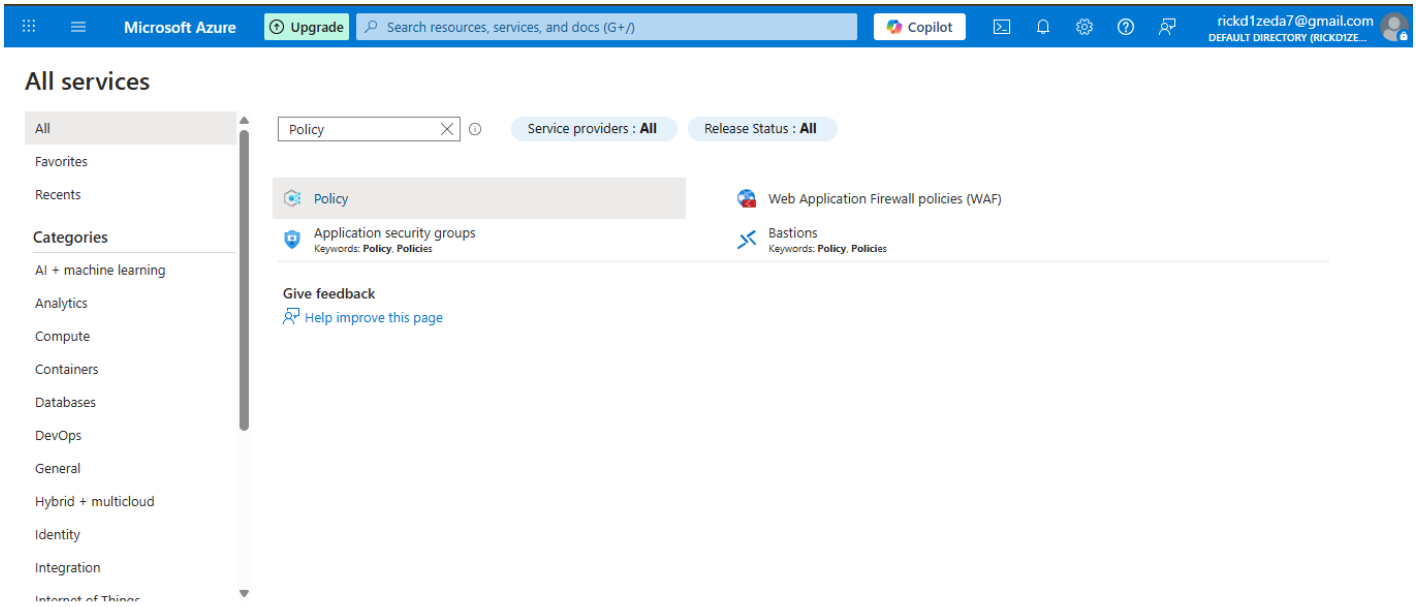
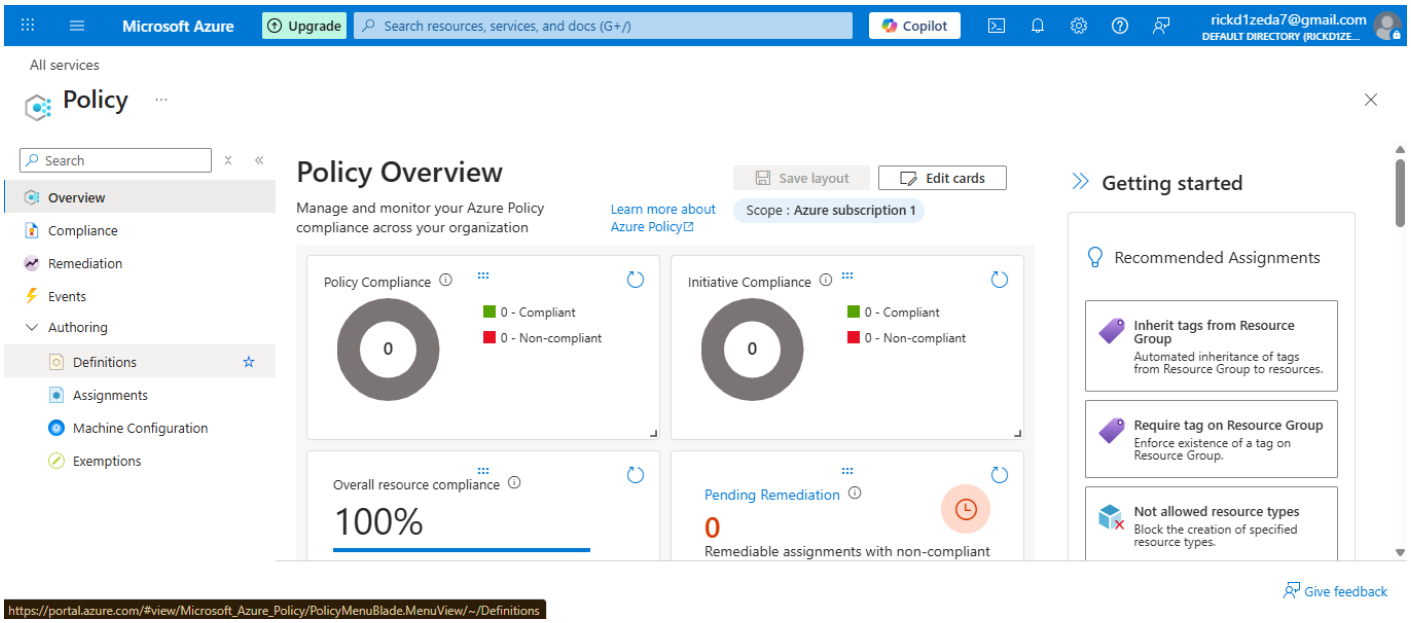
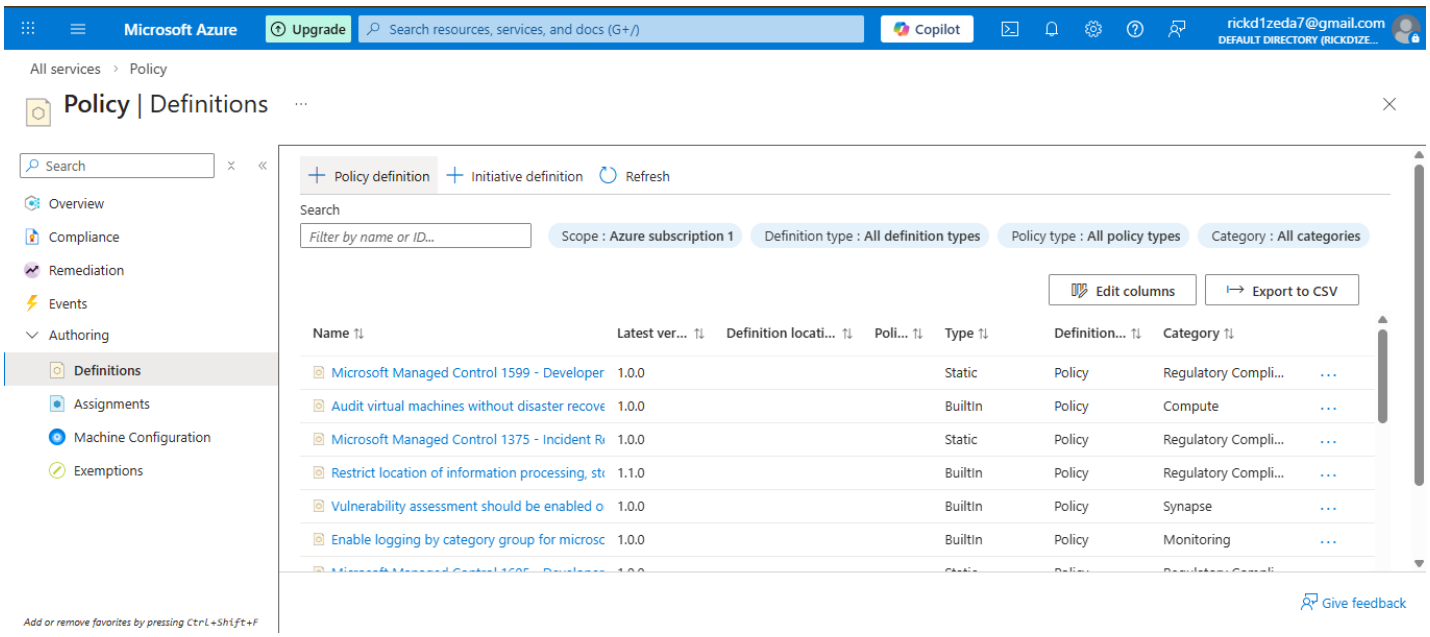
Lab Activity 3: Creating a custom Policy



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All services > Policy | Definitions

## Policy definition

New Policy definition

Basics

Definition location * : Azure subscription 1

Name * : DenyG-series

Description : Deny users from using certain virtual machines

Category : Create new / Use existing

Humanresources

Save | Give feedback

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All services > Policy | Definitions

## Policy definition

New Policy definition

Basics

```
1  {
2    "mode": "All",
3    "policyRule": {
4      "if": {
5        "not": {
6          "field": "location",
7          "in": "[parameters('allowedLocations')]"
8        }
9      },
10     "then": {
11       "effect": "audit"
12     }
13   },
14   "parameters": {
```

Invalid JSON

Save | Give feedback

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All services > Policy | Definitions

## DenyG-series

Policy definition

Assign policy | Edit definition | Duplicate definition | Delete definition

Essentials

| | |
|---|---|
| Name<br>DenyG-series | Definition location<br>Azure subscription 1 |
| Description<br>Deny users from using certain virtual machines | Definition ID<br>/subscriptions/3db70e8c-3a88-4f7e-b410-dd2d0a0b0d38/pr... |
| Available Effects<br>Audit | Type<br>Custom |
| Category<br>Humanresources | Mode<br>All |

Definition | Assignments | Parameters (1)

```
1  {
2    "properties": {
3      "displayName": "DenyG-series"
```

Give feedback

https://portal.azure.com/#blade/Microsoft_Azure_ActivityLog/ActivityLogBlade/queryInputs/%7B"user"%3A"%40me"%7D

## Notifications

More events in the activity log → | Dismiss all

**Creating policy definition succeeded**

Creating policy definition 'DenyG-series' in 'Azure subscription 1' was successful.

a few seconds ago

**Resource group created**

Creating resource group 'Developers-RG' in subscription 'Azure subscription 1' succeeded.

Go to resource group | Pin to dashboard

an hour ago

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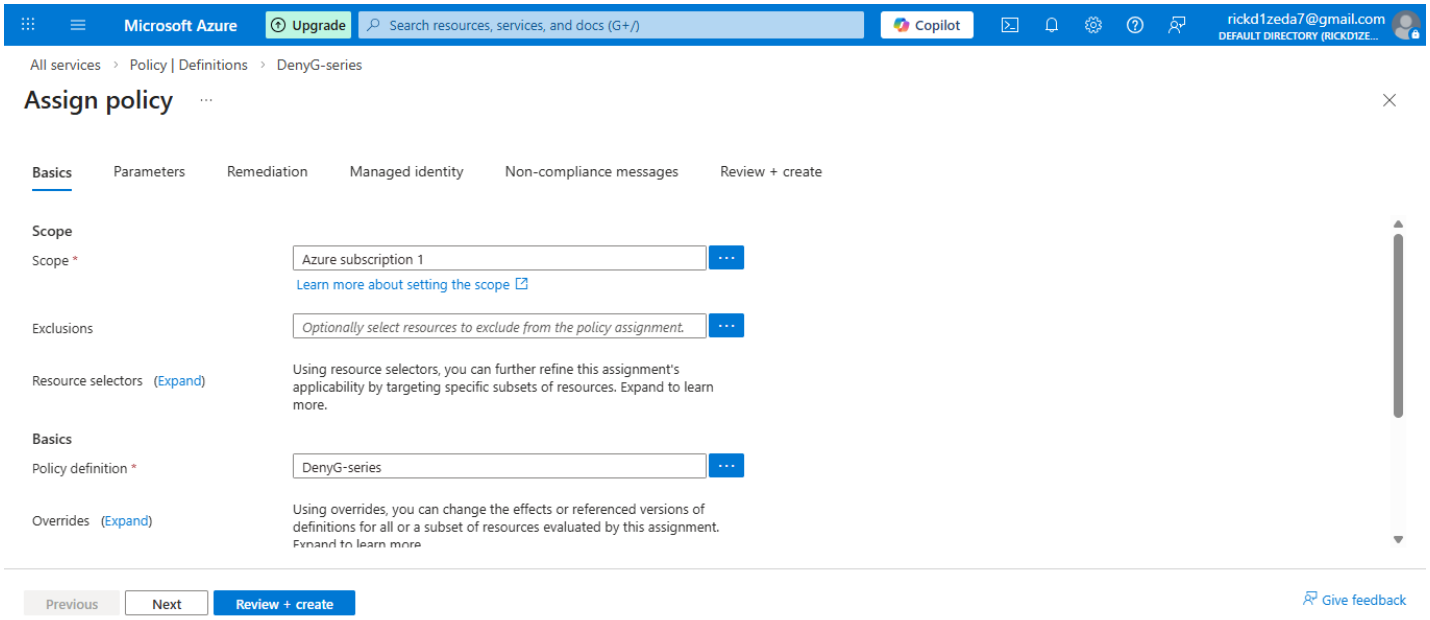


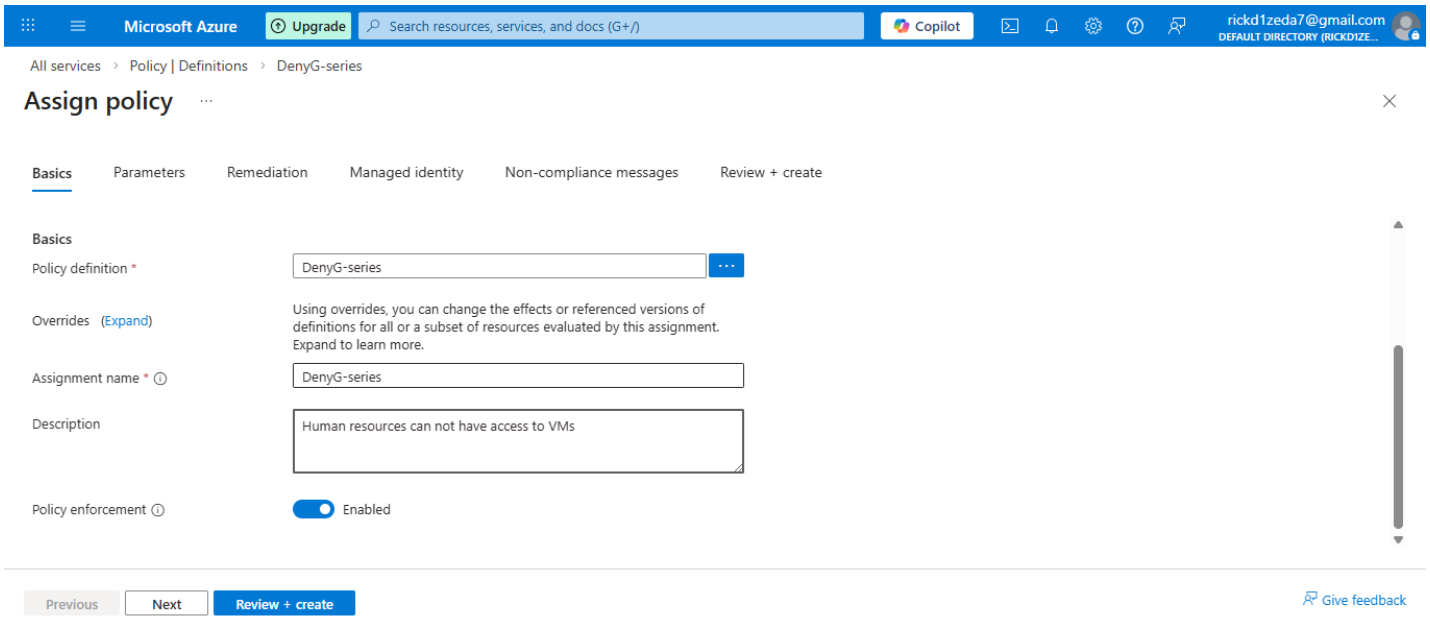


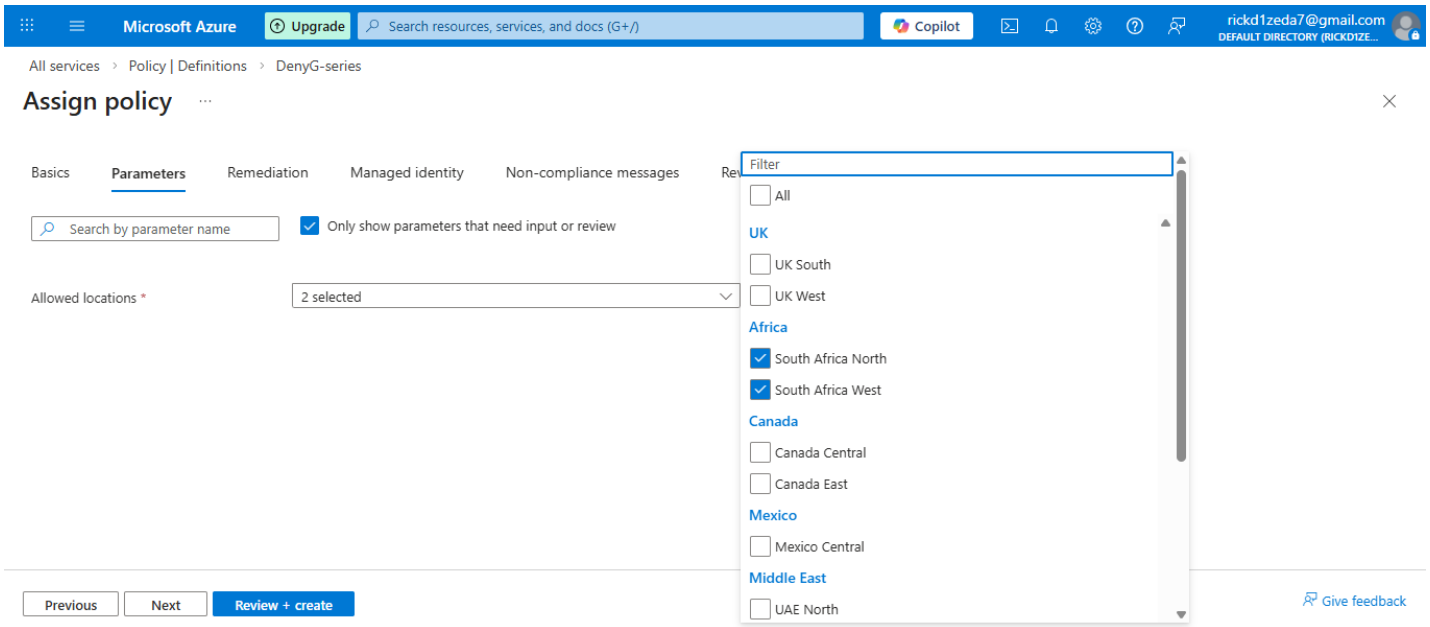

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All services > Policy | Definitions > DenyG-series

## Assign policy

Basics | Parameters | Remediation | Managed identity | Non-compliance messages | Review + create

### Basics

| | |
|---|---|
| Scope | Azure subscription 1 |
| Exclusions | -- |
| Policy definition | DenyG-series |
| Assignment name | DenyG-series |
| Description | Human resources can not have access to VMs |
| Policy enforcement | Default |
| Assigned by | Tony Makwanyane |

### Advanced

| | |
|---|---|
| Resource selectors | No selectors associated with this assignment. |
| Overrides | No overrides associated with this assignment |

Previous | Next | Create | Give feedback

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All services > Policy | Definitions

## DenyG-series

Policy definition

Assign policy | Edit definition | Duplicate definition | Delete definition

### Essentials

| | |
|---|---|
| Name<br>DenyG-series | Definition location<br>Azure subscription 1 |
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| Available Effects<br>Audit | Type<br>Custom |
| Category<br>Humanresources | Mode<br>All |

Definition | Assignments | Parameters (1)

```
46  "createdAt": "2026-03-08T19:56:52.7228946Z",
47  "lastModifiedBy": "rickd1zeda7@gmail.com",
```

Give feedback

### Notifications

More events in the activity log → | Dismiss all

**Creating policy assignment succeeded**

Creating policy assignment 'DenyG-series' in 'Azure subscription 1' was successful. Please note that the assignment takes around 5-15 minutes to take effect.

a few seconds ago

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All services > Policy

## Policy | Assignments

Search

- Overview
- Compliance
- Remediation
- Events
- Authoring
  - Definitions
  - Assignments
  - Machine Configuration
  - Exemptions

Assign policy | Assign initiative | Refresh

Search: Filter by name or ID...

Scope : Azure subscription 1 | Definition type : All definition types

| Total Assignments | Initiative Assignments | Policy Assignments |
|---|---|---|
| 1 | 0 | 1 |

Edit columns | Export to CSV

| Assignment name | Scope | Type |
|---|---|---|
| DenyG-series | Azure subscription 1 | Policy |

Give feedback

Add or remove favorites by pressing Ctrl+Shift+F

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20241063
Short report: The first lab activity was very simple to execute and is something I easily grasped. The second lab of creating role-based access control opened my eyes to how many permissions are available. The last lab was a little bit complex compared to the first two, but I showed standard of not only security but of auditing in Azure.